

Intro to System Dynamics

EGR 345

GVSU

Outline

- what is this class about?
 - what is a dynamic system?
 - what is feedback control?
 - why do these things matter?
- who am I?
- who are you?
- my teaching philosophy and the importance of this course?

What is a dynamic system?

- a system whose transient response matters
 - when the input changes, it does not immediately go to the new output
 - something happens before it settles at the new output state
- a system whose input/output relationship involves differential equations

What is feedback control?

Controls engineering in west Michigan

- what is a controls engineering in west Michigan?
 - what does that have to do with this class?

Is this class really important?

Will you learn more...

- will you learn more if you believe this course is important?
- will you learn more if you understand my teaching philosophy?